

MSSQL

```

/* ===== DATA CONTROL LANGUAGE (DCL)
LAB   Microsoft SQL Server (MSSQL)   Beginner to Intermediate Level
eCommerce Security Use Case===== */
-- =====

```

```
STEP 1: CREATE DATABASE-- =====
```

```
CREATE DATABASE ecommerce_security_db;GO
USE ecommerce_security_db;GO
```

```
/* =====
```

```
STEP 2: CREATE SAMPLE TABLES===== */
```

```
CREATE TABLE products (
    product_id INT IDENTITY(1,1) PRIMARY KEY,
    product_name VARCHAR(100),
    price DECIMAL(10,2),
    stock_qty INT
);GO
```

```
CREATE TABLE orders (
```

```
    order_id INT IDENTITY(1,1) PRIMARY KEY,
    customer_name VARCHAR(100),
    total_amount DECIMAL(10,2)
);GO
```

```
/* =====
```

```
STEP 3: INSERT SAMPLE DATA===== */
INSERT INTO products(product_name, price, stock_qty)VALUES('Laptop', 1200000,
10),('Mouse', 25000, 100),('Keyboard', 45000, 50),('Monitor', 350000,
20),('Printer', 450000, 5);GO
INSERT INTO orders(customer_name, total_amount)VALUES('Aung Aung',
1250000),('Su Su', 45000),('Kyaw Kyaw', 300000);GO
```

```

68  /* =====
69  STEP 4: VIEW SAMPLE DATA
70  ===== */
71
72  SELECT * FROM products;
73  GO

```

99 % ● No issues found Ln: 73, Ch: 3 SPC CRLF UTF-8

product_id	product_name	price	stock_qty
1	Laptop	1200000.00	10
2	Mouse	25000.00	100
3	Keyboard	45000.00	50
4	Monitor	350000.00	20
5	Printer	450000.00	5

```

74
75  SELECT * FROM orders;
76  GO

```

99 % ● No issues found Ln: 76, Ch: 3 SPC CRLF UTF-8

order_id	customer_name	total_amount
1	Aung Aung	1250000.00
2	Su Su	45000.00
3	Kyaw Kyaw	300000.00

```

/* =====
STEP 5: CREATE SQL SERVER LOGINS NOTE: Logins are created at SQL Server
level===== */

```

```

CREATE LOGIN admin_userWITH PASSWORD = 'Admin@123';GO
CREATE LOGIN sales_userWITH PASSWORD = 'Sales@123';GO
CREATE LOGIN report_userWITH PASSWORD = 'Report@123';GO

```

```
/* =====
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STEP 6: CREATE DATABASE USERS NOTE: Users are mapped to logins===== */

```
USE ecommerce_security_db;GO
CREATE USER admin_userFOR LOGIN admin_user;GO
CREATE USER sales_userFOR LOGIN sales_user;GO
CREATE USER report_userFOR LOGIN report_user;GO
```

```
117 -- check user roles
118
119 SELECT
120     dp1.name AS database_role,
121     dp2.name AS database_user
122 FROM sys.database_role_members drm
123 JOIN sys.database_principals dp1
124     ON drm.role_principal_id = dp1.principal_id
125 JOIN sys.database_principals dp2
126     ON drm.member_principal_id = dp2.principal_id
127 WHERE dp2.name = 'report_user';
128
```

99 % No issues found Ln: 118, Ch: 1 SPC CRLF UTF-8

database_role	database_user
report_user	report_user

```
128
129 /* =====
130 STEP 7: VIEW USERS
131 ===== */
132
133 SELECT
134     name,
135     type_desc
136 FROM sys.database_principals
137 WHERE type IN ('S', 'U');
138 GO
```

99 % No issues found Ln: 127, Ch: 32 SPC CRLF UTF-8

name	type_desc
dbo	WINDOWS_USER
guest	SQL_USER
INF	SQL_USER
sys	SQL_USER
adm	SQL_USER
sale	SQL_USER
repo	SQL_USER

```
-- =====
ADMIN USER-- Full control on database--
=====
GRANT CONTROLON DATABASE::ecommerce_security_dbTO admin_user;GO
```

```
/* =====
SALES USER Can: - Read products - Read orders - Insert
orders===== */
GRANT SELECTON productsTO sales_user;GO
GRANT SELECT, INSERTON ordersTO sales_user;GO
```

```
/* =====
REPORT USER Read-only access===== */
GRANT SELECTON productsTO report_user;GO
GRANT SELECTON ordersTO report_user;GO
```

```
187 /* =====
188 STEP 9: TEST admin_user
189 ===== */
190
191 EXECUTE AS USER = 'admin_user';
192 GO
193
194 SELECT * FROM products;
195 GO
196
```

99 % No issues found Ln: 192, Ch: 3 SPC CRLF UTF-8

product_id	product_name	price	stock_qty
1	Laptop	1200000.00	10
2	Mouse	25000.00	100
3	Keyboard	45000.00	50
4	Monitor	350000.00	20
5	Printer	450000.00	5

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197 SELECT * FROM orders;
198 GO

```

99 % No issues found Ln: 192, Ch: 3 SPC CRLF UTF-8

order_id	customer_name	total_amount
1	Aung Aung	1250000.00
2	Su Su	450000.00
3	Kyaw Kyaw	3000000.00

```

200 REVERT;
201 GO

```

99 % No issues found Ln: 200, Ch: 8 SPC CRLF UTF-8

order_id	customer_name	total_amount
1	Aung Aung	1250000.00
2	Su Su	450000.00
3	Kyaw Kyaw	3000000.00
4	Mya Mya	5500000.00

```

202
203 /* =====
204 STEP 10: TEST sales_user===== */
205
206 EXECUTE AS USER = 'sales_user';
207 GO
208
209 -- Allowed: View products
210
211 SELECT * FROM products;
212 GO
213

```

99 % No issues found Ln: 200, Ch: 8 SPC CRLF UTF-8

product_id	product_name	price	stock_qty
1	Laptop	1200000.00	10
2	Mouse	250000.00	100
3	Keyboard	450000.00	50
4	Monitor	3500000.00	20
5	Printer	4500000.00	5

```

220 -- Allowed: Insert orders
221
222 INSERT INTO orders
223 (customer_name, total_amount)
224 VALUES
225 ('Mya Mya', 5500000);
226 GO
227
228 SELECT * FROM orders;
229 GO
230
231 REVERT;

```

99 % No issues found Ln: 226, Ch: 3 SPC CRLF UTF-8

order_id	customer_name	total_amount
1	Aung Aung	1250000.00
2	Su Su	450000.00
3	Kyaw Kyaw	3000000.00
4	Mya Mya	5500000.00

```

/* =====
STEP 11: TEST report_user===== */
EXECUTE AS USER = 'report_user';GO
-- Allowed: Read-only access
SELECT * FROM products;GO
SELECT * FROM orders;GO
REVERT;GO
SELECT USER_NAME();

/* =====
STEP 12: TEST report_user CANNOT
INSERT===== */
EXECUTE AS USER = 'report_user';GO
-- This should FAIL
INSERT INTO orders(customer_name, total_amount)VALUES('Unauthorized User',
1000);GO

-- REVERT back to dbo/admin
REVERT;GO

/* =====
EXPECTED ERROR: INSERT permission
denied===== */

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/* =====
STEP 13: REVOKE PERMISSION  Remove INSERT permission from
sales_user===== */
REVOKE INSERTON ordersFROM sales_user;GO

/* =====
STEP 14: TEST REVOKE===== */
EXECUTE AS USER = 'sales_user';GO
-- This should FAIL after REVOKE
INSERT INTO orders(customer_name, total_amount)VALUES('Blocked User', 5000);GO
REVERT;GO

/* =====
STEP 15: DENY DELETE PERMISSION===== */
DENY DELETEON ordersTO sales_user;GO

/* =====
STEP 16: TEST DENY===== */
EXECUTE AS USER = 'sales_user';GO

-- This should FAIL
DELETE FROM ordersWHERE order_id = 1;GO
REVERT;GO

```

```

330  /* =====
331  STEP 17: VIEW USER PERMISSIONS
332  ===== */
333
334  SELECT
335
336      pr.name AS user_name,
337
338      pe.permission_name,
339
340      pe.state_desc,
341
342      ob.name AS object_name
343
344  FROM sys.database_permissions pe
345
346  JOIN sys.database_principals pr
347      ON pe.grantee_principal_id = pr.principal_id
348
349  JOIN sys.objects ob
350      ON pe.major_id = ob.object_id
351
352  ORDER BY pr.name;
353  GO

```

99 % 99 % No issues found Ln: 332, Ch: 30 SPC CRLF UTF-8

Results Messages

	user_name	permission_name	state_desc	object_name
1	report_user	SELECT	GRANT	products
2	report_user	SELECT	GRANT	orders
3	sales_user	SELECT	GRANT	products
4	sales_user	DELETE	DENY	orders
5	sales_user	SELECT	GRANT	orders

```

355  /* =====
356  STEP 18: VIEW CURRENT SESSION USER
357  ===== */
358
359  SELECT
360      USER_NAME() AS current_database_user;
361  GO
362

```

99 % 99 % No issues found Ln: 332, Ch: 30 SPC CRLF UTF-8

Results Messages

	current_database_user
1	dbo

```

362  /* =====
363  STEP 19: VIEW CURRENT LOGIN
364  ===== */
365
366  SELECT
367      SYSTEM_USER AS current_login;
368  GO
369
370

```

99 % 99 % No issues found Ln: 332, Ch: 30 SPC CRLF UTF-8

Results Messages

	current_login
1	LINEAPHYU\Ausp

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371  /* =====
372  STEP 20: SHOW LOGIN INFORMATION
373  ===== */
374
375  SELECT
376      name,
377      type_desc,
378      create_date
379  FROM sys.server_principals
380  WHERE type IN ('S', 'U');
381  GO

```

99 % No issues found Ln: 332, Ch: 30 SPC CRLF UTF-8

	name	type_desc	create_date
5	NT SERVICE\Winmgmt	WINDOWS...	2026-05-10 20:35:41.793
6	NT Service\MSSQL\$SQLEXPRESS	WINDOWS...	2026-05-10 20:35:41.800
7	NT AUTHORITY\SYSTEM	WINDOWS...	2026-05-10 20:35:41.813
8	NT SERVICE\SQLTELEMETRYSS...	WINDOWS...	2026-05-10 20:35:42.673
9	##MS_PolicyEventProcessingLogi...	SQL_LOGIN	2026-05-15 22:44:06.363
10	admin_user	SQL_LOGIN	2026-05-17 16:52:15.633
11	sales_user	SQL_LOGIN	2026-05-17 16:52:15.733
12	report_user	SQL_LOGIN	2026-05-17 16:52:15.737

```

/* =====
STEP 21: CLEANUP SECTION  DROP
USERS===== */
DROP USER admin_user;GO
DROP USER sales_user;GO
DROP USER report_user;GO

/* =====
STEP 22: DROP LOGINS===== */
DROP LOGIN admin_user;GO
DROP LOGIN sales_user;GO
DROP LOGIN report_user;GO

/* =====
END OF DCL LAB SCRIPT
===== */

```